Yendluri Ajay

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.60
NARAYANA JUNIOUR COLLEGE	Class XII (BIEAP)	2019	9.86%
NARAYANA HIGH SCHOOL	Class X (BSEAP)	2017	9.50%

MTECH THESIS

Progressive Damage and Failure Analysis of Open-Hole Variable Angle Tow CFRP Composites

- Developed an Integrated LaRC05 intralaminar criteria, non-linear shear degradation, and Cohesive Zone Modeling (CZM) in Abaqus to simulate coupled fiber, matrix, and delamination failure.
- Fiber Path Optimization: Optimized curvilinear fiber path steering around open holes to achieve continuous placement, eliminating stress concentrations and increasing load capacity.
- Experimental Validation: Fabricated CFRP coupons on an automated TFP machine and to be tested under tensile loading and validate against computational models using DIC and Acoustic Emission (AE) damage tracking.

PROJECTS

Design & Failure Analysis of a 700-Bar Type IV Hydrogen Composite Pressure Vessel

- Sized a 60L tank (UN R134 / ISO 11119-3) via Python netting analysis and modeled variable winding angles across PA6 liner and T700 overwrap in ANSYS ACP.
- Conducted Hashin Progressive Failure Analysis, predicting a 161.2 MPa burst pressure (2.30x NWP) at 5.6 wt% gravimetric storage density.

ML Pipeline for EV Chassis Material Selection

- Developed an automated Python pipeline to clean data for 1,400+ engineering materials, creating custom strength-to-weight metrics and isolating key mechanical properties for EV chassis components.
- Built and hyperparameter-tuned a Scikit-Learn Neural Network classifier, achieving 93% accuracy in predicting material suitability ratings for optimal lightweight alloys.

Aerodynamic Design & Flight Testing of Fixed-Wing Gliders

- Fabricated and flight-tested conventional and canard gliders, optimizing airfoil profiles, CG location, and static stability through experimental flight trials.

Design, CFD Simulation, and Field Testing of a Cased Banki Hydrokinetic Turbine

- Designed, simulated (ANSYS Fluent), and 3D-printed an 8-blade vertical-axis turbine with a custom flow-deflector casing, achieving self-starting capability and 14.34% peak efficiency in low-velocity (0.40-0.455 m/s) canal currents

SKILLS

CAD & FEA: SolidWorks, AutoCAD, Abaqus, ANSYS Mechanical, ANSYS Fluent

Composite Manufacturing: Tailored Fiber Placement (TFP), Autoclave Processing, Hand Lay-up, Prepreg Lay-up.

Experimental Mechanics & NDT: Material Characterization as per ASTM standards, Mechanical Testing, Ultrasonic C-Scan, Digital Image Correlation (DIC), Acoustic Emission (AE)

Programming & Numerical Computing: MATLAB, Python, Fortran

Machine Learning & Data Analysis: Machine Learning Fundamentals, Data Processing, Numerical Data Analysis

RELEVANT COURSES

Analysis and Design of Composite Structures | Finite Element Method (FEM) | Machine Learning & its Applications |

Aircraft Structures & Dynamics | Aerodynamics & Propulsion | High-Speed Aerodynamics

INTERNSHIP

KSRTC | Regional workshop, Aluva

- Gained hands-on exposure to commercial vehicle subsystems, including diesel engine powerplants, transmission drives, steering, suspension, and pneumatic braking systems.
- Studied preventive and corrective maintenance workflows, vehicle inspections, and diagnostic protocols to improve fleet reliability and operational efficiency.

POSITION OF RESPONSIBILITY

Teaching Assistant - Machine Learning and Its Applications

Evaluated course assignments and corrected examination papers.

HOBBIES AND INTERESTS

- Enjoys playing cricket,cycling and travelling, with a keen interest in exploring new places and cultures.
- Passionate about reading magazines and journals on composite materials and keeping up with emerging technologies and engineering innovations.